

Question ID: e2eb70b9

Question

While researching a topic, a student has taken the following notes:

- Traditionally, manufacturers have dyed denim jeans blue by dipping them in a solution containing indigo powder.
- Indigo doesn't dissolve in just water, so manufacturers must mix hazardous chemicals with water to dissolve the powder.
- Textile researcher Smriti Rai discovered a process for dyeing blue jeans without chemicals.
- Rai added indigo powder to a hydrogel containing nanocellulose and produced a dye that could be spread directly onto the denim.
- Nanocellulose is a natural, plant-based substance that separates the molecules of indigo powder.

The student wants to emphasize a difference between the two approaches to dyeing blue jeans. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. Though created using a different process, Rai's dye contains the same ingredient as the dye produced by blue jean manufacturers.
- B. Nanocellulose is a natural, plant-based substance that separates the molecules of indigo powder, which doesn't dissolve in water.
- C. The traditional approach to dyeing blue jeans is to dip them in a solution containing hazardous chemicals.
- D. Rai's approach substitutes a natural, plant-based substance for the hazardous chemicals that manufacturers have traditionally used.